

5-year integrated M.Sc. (Computer science)
Semester-V
Java Programming
Assignment-I

1. Define UNICODE.
2. Define Byte code.
3. Explain JDK root directory with diagram.
4. Discuss five keywords of Java with example.
5. Discuss the diamond problem with reference to inheritance.
6. Difference between finally, finalize and final keyword.
7. Discuss the generation of languages.
8. How is platform independence achieved in Java?
9. Explain: Class, Encapsulation, Inheritance and Polymorphism.
10. List the commonly found tools available as part of JDK which are not part of JRE?
11. Difference between primitive and reference types.
12. Discuss two types of byte ordering with example. Mention the difference between the two.
13. Why should we have util package?
14. Explain Method Overloading in Java.
15. Explain Constructors in Java.
16. Explain static variables and static methods in Java.
17. Discuss the purpose of final keyword and its advantages.
18. Discuss finalize method and its importance in garbage collection in Java.
19. Define instance/member variables.
20. Define functional programming.
21. Why was Lambda expressions introduced?
22. Discuss Lambda expressions and its implementation in Java.

23. Discuss the benefit of generic over non-generic types.
24. Explain Thread Life-cycle with diagram.
25. Compare concurrency and parallelism with reference to multithreading in Java.
26. Difference between multitasking and multithreading with reference to threads in Java.
27. What are threads good for?
28. Discuss Application Thread with example.
29. When should we extend Thread class and when should we implement Runnable interface in Java?
30. Difference between isAlive() and join() in Java.
31. Explain immutable objects with reference to Threads.
32. Difference between '==' and equals() in Java.
33. Compare String and StringBuilder class with methods.
34. Compare boxing and unboxing in Java. Also define auto boxing.
35. Difference between Byte Streams and Character Streams.
36. Discuss any five byte stream classes.
37. Discuss predefined streams of Java.
38. Difference between channel and heap.
39. Discuss situations where String Tokenizer class is used.
40. Discuss Exception and types of Exceptions in Java.
41. Differentiate Interface and Polymorphism.
42. Differentiate Multiple Inheritance and Interface.
43. Discuss BinaryStream and CharacterStream.
44. Discuss three subclasses of Error Runtime Exception.
45. Discuss classes for input and output in Java.
46. Discuss methods for input and output in Java.
47. Explain seven different subclasses of InputStream.

48. Discuss methods of OutputStream class.
49. Discuss base classes for StreamReader and StreamWriter alongwith their methods. 50. How to customize a StringTokenizer?
50. Which tokens can a StringTokenizer recognize in its default state?
51. Explain Locale class.
52. Compare absolute path and relative path.
53. Discuss the methods of File object OR Explain reading and writing in files in Java.
54. Explain channels with reference to file-handling in Java.
55. Explain serializing objects in Java.
56. What are generic class types?
57. What are generic constructors?
58. Define Threads. Differentiate Daemon Threads and User Threads.
59. Discuss thread scheduling with example.
60. Explain Synchronization and Deadlocks in threading.
61. Discuss methods used in thread communication.
62. Explain thread priorities.
63. Discuss inter-process communication.